

Muhammad Maaz

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Google Scholar

Citations: 6500+



LinkedIn



GitHub

Research Interests

Multimodal Large Language Models (MLLMs) for in-the-wild image and video understanding, multimodal reasoning, high-resolution videos processing, and long-video understanding.

Education

- 2022 – 2026 **Ph.D. Computer Vision** Mohamed bin Zayed University of Artificial Intelligence.
Advisor: [Dr. Salman Khan](#)
- 2020 – 2022 **M.Sc. Computer Vision** Mohamed bin Zayed University of Artificial Intelligence.
Thesis Title: Class-agnostic Object Detection with Multi-modal Transformer.
Advisor: [Dr. Salman Khan](#)
- 2014 – 2018 **B.Sc. Electrical Engineering** University of Engineering and Technology.
Thesis Title: Intelligent Discipline Maintenance System with Client-Server Topology.
Advisor: [Dr. Kashif Javed](#)

Experience

- Mar. 2025 – Aug. 2025 **Research Scientist Intern (Internship # 2) @ Meta – San Francisco**
Conducted research on incorporating promptable image segmentation into Multimodal Large Language Models (MLLMs) to improve fine-grained visual reasoning.
- Jun. 2024 – Dec. 2025 **Research Scientist Intern (Internship # 1) @ Meta – San Francisco**
Developed Perception Language Model (PLM), a family of open and fully reproducible multimodal large language models (MLLMs). PLM matches or exceeds the performance of recent state-of-the-art MLLMs such as InternVL3 and Qwen2.5VL, using fully open data.
- Jul. 2020 – Dec. 2020 **Computer Vision Engineer, Hazen.ai – Lahore**
Developed a traffic light phase detection solution for road safety applications. Trained a neural network using triplet loss to learn robust embeddings for different traffic light phases.
- Jun. 2018 – Jul. 2020 **Software Engineer, Confiz Limited – Lahore**
Developed a person detection and tracking system to identify engaged and ignored areas in retail stores, and a face recognition system to generate visitor demographics and visit frequencies.

Research

- Under Review **Video-R2: Reinforcing Consistent and Grounded Reasoning in Multimodal Language Models**
[Muhammad Maaz](#), Hanoona Rasheed, Salman Khan, Fahad Khan
| [Paper](#) | [Code](#) | [Hugging Face](#) |
- Under Review **Video-CoM: Interactive Video Reasoning via Chain of Manipulations**
Hanoona Rasheed, Mohammed Zumri, [Muhammad Maaz](#), Ming-Hsuan Yang, Fahad Khan, Salman Khan
| [Paper](#) | [Code](#) |

Research (continued)

NeurIPS 2025

Spotlight

2100+ ☆

40K+ Monthly 

PerceptionLM: Open-Access Data and Models for Detailed Visual Understanding

Muhammad Maaz*, Jang Hyun Cho*, Andrea Madotto*, Effrosyni Mavroudi*, Triantafyllos Afouras*, Tushar Nagarajan*, Yale Song*, Tengyu Ma*, Shuming Hu*, Suyog Jain, Miguel Martin, Huiyu Wang, Hanoona Rasheed, Peize Sun, Po-Yao Huang, Daniel Bolya, Nikhila Ravi, Shashank Jain, Tammy Stark, Shane Moon, Babak Damavandi, Vivian Lee, Andrew Westbury, Salman Khan, Philipp Krähenbühl, Piotr Dollár, Lorenzo Torresani, Kristen Grauman, Christoph Feichtenhofer

| [Paper](#) | [Code](#) | [Hugging Face](#) |

ACL 2024

1350+ ☆

1600+ Citations

Video-ChatGPT: Towards Detailed Video Understanding via Large Vision and Language Models

Muhammad Maaz*, Hanoona Rasheed*, Salman Khan, Fahad Khan

| [Paper](#) | [Code](#) | [Hugging Face](#) | [Benchmark](#) |

arXiv Preprint

270+ ☆

110+ Citations

VideoGPT+: Integrating Image and Video Encoders for Enhanced Video Understanding

Muhammad Maaz, Hanoona Rasheed, Salman Khan, Fahad Khan

| [Paper](#) | [Code](#) | [Hugging Face](#) |

CVPR 2024

860+ ☆

530+ Citations

GLaMM: Pixel Grounding Large Multimodal Model

Hanoona Rasheed*, **Muhammad Maaz***, Sahal Shaji Mullappilly, Abdelrahman Shaker, Salman Khan, Hisham Cholakkal, Rao M. Anwer, Erix Xing, Ming-Hsuan Yang, Fahad Khan

| [Paper](#) | [Code](#) | [Hugging Face](#) | [Dataset](#) | [Website](#) |

ICLR 2026

VideoMathQA: Benchmarking Mathematical Reasoning via Multimodal Understanding in Videos

Hanoona Rasheed, Abdelrahman Shaker, Anqi Tang, **Muhammad Maaz**, Ming-Hsuan Yang, Salman Khan, Fahad Khan

| [Paper](#) | [Website](#) | [Hugging Face](#) |

arXiv Preprint

Mobile-VideoGPT: Fast & Accurate Video Understanding Language Model

Abdelrahman Shaker, **Muhammad Maaz**, Chenhui Gou, Hamid Reza Tofighi, Salman Khan, Fahad Khan

| [Paper](#) | [Code](#) | [Hugging Face](#) |

WACV 2025

Oral

PALO: A Polyglot Large Multimodal Model for 5B People

Hanoona Rasheed, **Muhammad Maaz**, Abdelrahman Shaker, Salman Khan, Hisham Cholakkal, Rao M. Anwer, Tim Baldwin, Michael Felsberg, Fahad Khan

| [Paper](#) | [Code](#) | [Dataset](#) |

CVPR 2023

270+ ☆

330+ Citations

Fine-tuned CLIP Models are Efficient Video Learners

Hanoona Rasheed, Muhammad Uzair Khattak, **Muhammad Maaz**, Salman Khan, Fahad Khan

| [Paper](#) | [Code](#) | [Website](#) |

CVPR 2023

740+ ☆

1680+ Citations

MaPLE: Multi-modal Prompt Learning

Muhammad Uzair Khattak, Hanoona Rasheed, **Muhammad Maaz**, Salman Khan, Fahad Khan

| [Paper](#) | [Code](#) | [Website](#) |

ECCV 2022

310+ ☆

160+ Citations

Class-agnostic Object Detection with Multi-modal Transformer

Muhammad Maaz*, Hanoona Rasheed*, Salman Khan, Fahad Khan, Rao M. Anwer, Ming-Hsuan Yang

| [Paper](#) | [Code](#) | [Presentation](#) |

NeurIPS 2022

290+ ☆

230+ Citations

Bridging the Gap between Object and Image-level Representations for Open-Vocabulary Detection

Hanoona Rasheed*, **Muhammad Maaz***, Muhammad Uzair Khattak, Salman Khan, Fahad Khan

| [Paper](#) | [Code](#) | [Website](#) |

ECCVW 2022

Oral

370+ ☆

520+ Citations

EdgeNeXt: Efficiently Amalgamated CNN-Transformer Architecture for Mobile Vision Applications

Muhammad Maaz*, Abdelrahman Shaker*, Hisham Cholakkal, Salman Khan, Syed Waqas Zamir, Rao M. Anwer, Fahad Khan

| [Paper](#) | [Code](#) | [Website](#) |

Research (continued)

ICCV 2023 275+ ☆ 360+ Citations	SwiftFormer: Efficient Additive Attention for Transformer-based Real-time Mobile Vision Applications Abdelrahman Shaker, Muhammad Maaz , Hanoona Rasheed, Salman Khan, Ming-Hsuan Yang, Fahad Khan Paper Code
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Awards

AI Rising Star 2026	AI Rising Star Award at KAUST AI Rising Star Symposium 2026. Recognized for emerging research impact and contributions to the field of AI.
Graduate Research Excellence Award 2026	Recipient of the Graduate Research Excellence Award at MBZUAI's 5th Anniversary celebration, awarded as the only student honoree.
Google PhD Fellowship Award 2025	First in the MENA region and the first student in the UAE to receive the Google PhD Fellowship (2026), selected as one of 255 fellows worldwide from 35 countries, with the fellowship awarded for research in Machine Perception.

Invited Talks

KAUST AI Rising Star Symposium 2026	Invited for a Research Talk at the 5th edition of AI Rising Star Symposium at King Abdullah University of Science and Technology (KAUST).
IndabaX Sudan 2025	Invited to deliver a research talk at IndabaX Sudan 2025, the annual Machine Learning and AI conference and one of the largest gatherings in the African AI community.

Technical Skills

Distributed Computing	Hands-on experience with Deepspeed, FSDP, Model Parallelism (TP, PP, EP), Sequence Parallelism and Context Parallelism.
Others	MoE, MoT Architecture, Model training and evaluation at scale.

Community Contributions

2022 – Cont.	Serving as Reviewer Transformer Models in Vision, Special Issue in IEEE TPAMI, ECCVW 2022, CVPR 2023, ICCV 2023, ACCV 2024, ECCV 2024, ICCV 2025, NeurIPS 2025, ICLR 2026 and CVPR 2026.
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